

Vaccination Policy under Model Uncertainty

Can the Needs of the Few Outweigh the Needs of the Many?

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This talk asks when that intuition **fails**.

Outline

1 The Orthodoxy

2 The Claim

3 The Model

4 Results

5 Objection & Humility

6 Conclusion

The Orthodoxy

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We examine the evidential basis for “mass vaccination surely helps.”

The Claim

Our claim

There exist viral regimes in which vaccinating **only the vulnerable** produces **fewer total deaths** than vaccinating **everyone**, even when the vaccine is:

- safe (no adverse reactions, no nocebo),
- freely available in unlimited quantity,
- distributed and administered immediately,
- genuinely effective for the non-vulnerable.

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Two disclaimers, up front

This is **not** a claim about actual COVID-19, and **not** a policy recommendation. It is a **constructive counterexample** in service of **epistemic humility**.

The Model

An agent-based SISD model

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The acronym **SISD** names the states an agent moves through:

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- **Infected**: each step it either dies, recovers, or stays infected.
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In short: Susceptible $\rightarrow$ Infected $\rightarrow$ {recover (back to S, with immunity) **or** die}.

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“... a pathogen ... will be outcompeted by mutants that cause less damage to the host and allow it to ... infect other hosts.” [7, p. 71]

Two strategies, and why S2 might win

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So it is possible S2 yields fewer deaths, but only simulation can tell.

Results

Searching the parameter space

We fix most parameters and vary three: **viral ageing**, **vaccination effect**, **death probability**.

- Bayesian optimisation (`scikit-optimize`, GP regression) [8].
- Objective: average death difference $S1 - S2$, over $n = 200$ random seeds per setting.
- $k = 1000$ settings explored $\Rightarrow$ **224** where $S2$ beats $S1$.

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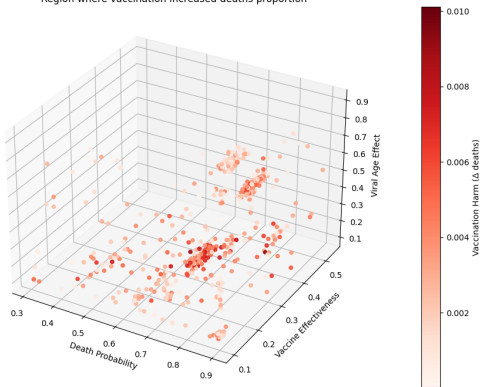
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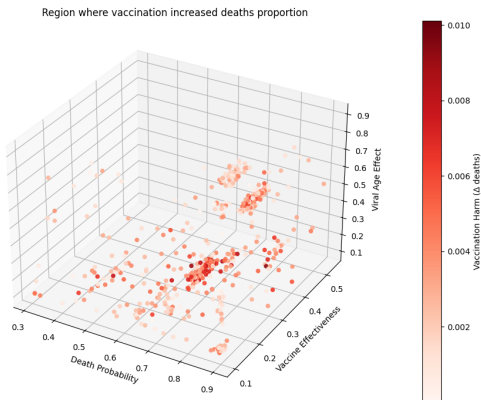
Goal: not that such regimes are common, but that they exist, and are spread out.

S2 causes fewer deaths than S1

Region where vaccination increased deaths proportion



S2 causes fewer deaths than S1



Two lessons:

- the advantage is **small** (max ~1% of agents on average), and
- the regions are **scattered**: no tidy rule for when S1 wins.

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Free-riding → public service

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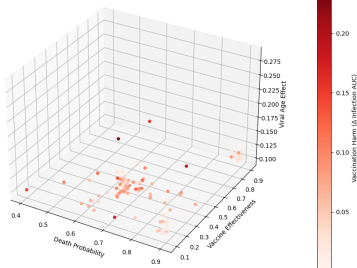
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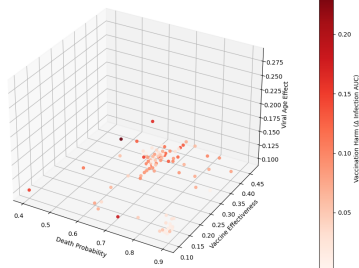
At times, the needs of the few **do** outweigh the needs of the many, if measured by simple averaging.

Beyond deaths: area under the curve

Region where vaccination increased infection AUC

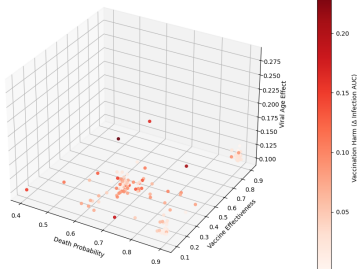


Region where vaccination increased both deaths proportion and infection AUC

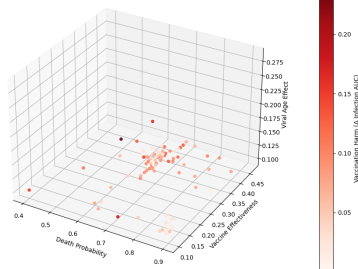


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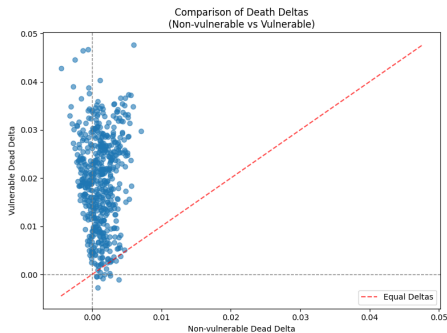


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Left: settings where S2 also has the smaller infection burden (an economic proxy).
Right: settings where S2 wins on **both** deaths and AUC. Fewer deaths $\neq$ smaller AUC.

Who gains from Strategy 2?

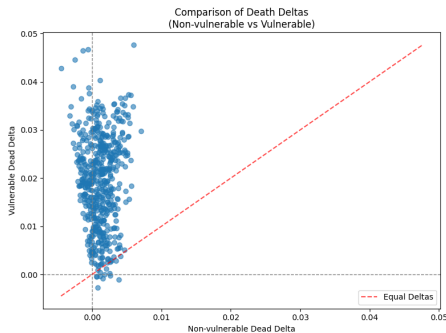


Each dot: one of the 224 settings.

x: S2 advantage for **non-vulnerable**.

y: S2 advantage for **vulnerable**.

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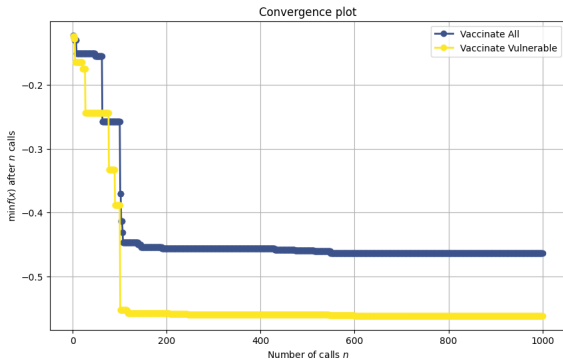
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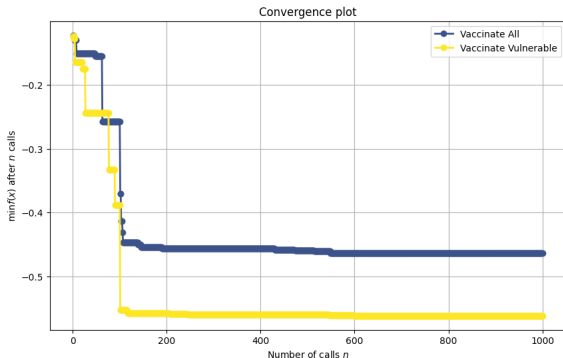
y : S2 advantage for **vulnerable**.

Top-left ($x < 0$, $y > 0$): the non-vulnerable majority **sacrifices**, the vulnerable minority gains more: net benefit overall.

Precautionary Principle: Worst cases cut the other way

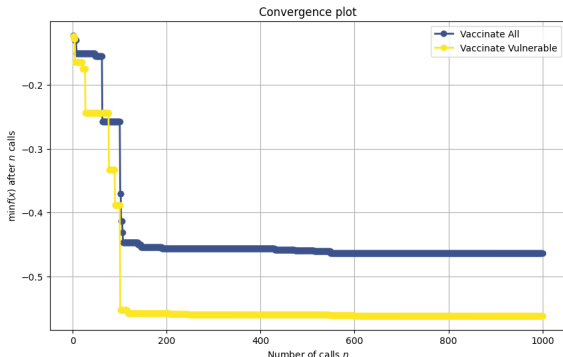


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The PP dictates that we may rationally prefer S1.

Objection & Humility

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Historical precedent: Marek's disease [10] and pneumococcal serotype replacement [11], vaccination with real negative evolutionary outcomes.

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Underdetermination

Even a coherent, empirically motivated model can leave the **ethical** conclusion substantially underdetermined.

Conclusion

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- We should be cautious before labelling vaccine hesitancy as free-riding, a boon, or negligence.
- Ethical **pluralism** only sharpens the point: different theories, different verdicts.

The remedy is epistemic humility and honest uncertainty communication, itself a contribution to pandemic preparedness.

Takeaways

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- 1 **A constructive counterexample:** under favourable assumptions, selective vaccination can still minimise deaths.
- 2 **A neglected mechanism:** the transmission-virulence trade-off (viral ageing) changes the ethical picture.
- 3 **A stance:** strong moral conclusions drawn straight from epidemiological models deserve humility under deep uncertainty.

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Questions?

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Appendix: Model parameters

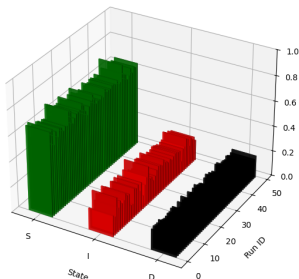
- $N = 500$ agents, 25×25 grid, 10% initially infected, 10% vulnerable.
- Base infection prob. 0.25; base death prob. 0.01; recovery time 30.
- Vulnerability penalty +300%; immunity effect 10%/recovery.
- Vaccination effect 70%; viral ageing 10%/transmission.
- Optimisation: $2 \times 200 \times 1000 = 400,000$ simulations, 11.5h on TPU.

Appendix: Death probability update

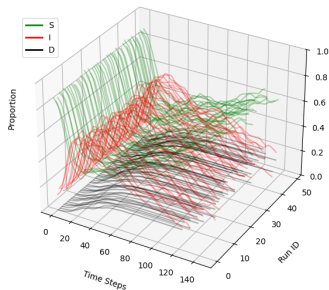
```
1 if self.vaxxed:
2     base_death_prob = self.death_prob * (1 - self.vax_effect)
3 else:
4     base_death_prob = self.death_prob
5
6 base_death_prob = max(
7     base_death_prob *
8     (1 - self.viral_age_effect * self.viral_age
9     - self.immune_adaptation_effect * self.immunity_level),
10    0)
11
12 if self.rng.random() < base_death_prob:
13     self.state = 'D'
```


Appendix: An example single run (S2)

Final Agent State Proportions (3D Bar)
Vaccinate Vulnerable



Smoothed State Proportions Over Time (3D)
Vaccinate Vulnerable



Susceptible (green), infected (red), dead (black). Virus becomes endemic but harmless via viral ageing and immunity growth.

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